Q1. BLP, Merger Simulation

1. Suppose two of the four firms were to merge. Give a brief intuition for what theory tells us is likely to happen to the equilibrium prices of each good j.

In this model, firms set price to compete for more shares. Especially in this case, two firms of the same products merge. Theoretically, after the merger, firms don't need to lower the price to get a bigger market share for the homogeneous products, thus the after-merger price will increase.

2. Suppose firms 1 and 2 are proposing to merge. Use the pyBLP merger simulation procedure to provide a prediction of the post-merger equilibrium prices

Table 1: Market Data with Merger Effects (Highlighted Post-Merger Prices)

Index	Price	Share	Market ID	Firm ID	Sat.	Wired	x	w	Merger ID	Post-Merger Price
0	2.764 784	0.086 328	0	0	1	0	0.253 759	0.717 270	1	3.045 445
1	2.643 251	0.200 549	0	1	1	0	0.134 024	0.000 864	1	2.764 323
2	2.827 237	0.227 401	0	2	0	1	0.140 025	1.290 341	2	2.834 349
3	2.603 836	0.133 281	0	3	0	1	0.107 216	0.368 745	3	2.607 697
4	3.008 030	0.397 030	1	0	1	0	1.560 624	1.403 483	1	3.078 016
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2395	2.707 730	0.013 148	598	3	0	1	0.414 766	0.915 487	3	2.708 995
2396	2.396 050	0.261 938	599	0	1	0	1.218 003	0.246 415	1	2.418 416
2397	3.720 164	0.011 428	599	1	1	0	0.954 761	2.082 937	1	4.166 963
2398	2.736 636	0.026 670	599	2	0	1	0.186 165	0.931 112	2	2.736 938
2399	3.372 414	0.553 740	599	3	0	1	2.295 622	0.049 518	3	3.380 117

Notes: Yellow highlighting indicates post-merger equilibrium prices. Sat./Wired are binary indicators (1=Yes).

3. Now suppose instead that firms 1 and 3 are the ones to merge. Re-run the merger simulation. Provide a table comparing the (average across markets) predicted merger-induced price changes for this merger and that in the previous item. Interpret the differences between the predictions for the two mergers.

Table 2: Average Prices by Firm and Scenario

Firm	Pre-Merger	Post-Merger 1-2	Post-Merger 1-3
1	2.73661748	3.01697727	2.84394956
2	2.72862259	3.00719153	2.73994762
3	2.76189030	2.77102592	2.86441443
4	2.75867146	2.76917149	2.77115566

The increase in price is bigger when two firms of homogeneous products merge. In the merger case of firm 1 and 2, where they both produce satellite TVs, the price goes from at around 2.70 to around 3.00. This is a 11% increase in price. Whereas in the case of firm of 1 and 4, where they produce different TVs, the price only goes to 2.8, a 3% increase in price. Merger with homogeneous products end up having more market powers.

4. Thus far you have assumed that there are no efficiencies (reduction in costs) resulting from the merger. Explain briefly why a merger-specific reduction in marginal cost could mean that a merger is welfare-enhancing.

Price is the marginal cost plus the mark-up. The mark-up is decided by the market power. When a two firms with homogeneous products merge, they have more market power as an entity, as the overall competition is gone. Thus, a mark-up is induced. When the merger brings down the marginal cost of production, the price goes down as a result, which offsets the upward pressure on the price due to the market power mark-up. The overall change is decided by which force dominates the other. The simulation in the following part suggests a 15% marginal cost reduction actually increase the total welfare.

5. Consider the merger between firms 1 and 2, and suppose the firms demonstrate that by merging they would reduce marginal cost of each of their products by 15%. Furthermore, suppose that they demonstrate that this cost reduction could not be achieved without merging. Using the pyBLP software, re-run the merger simulation with the 15% cost saving. Show the predicted post-merger price changes (again, for each product, averaged across markets). What is the predicted impact of the merger on consumer welfare, assuming that the total measure of consumers M_t is the same in each market t

Table 3: Average Prices by Firm and Scenario

Firm	Pre-Merger	Post-Merger 1-2	Post-Merger 1-2 New MC	Post-Merger 1-3
1	2.73661748	3.01697727	-0.08034946	2.84394956
2	2.72862259	3.00719153	-0.08002647	2.73994762
3	2.76189030	2.77102592	-0.00047972	2.86441443
4	2.75867146	2.76917149	-0.00161360	2.77115566

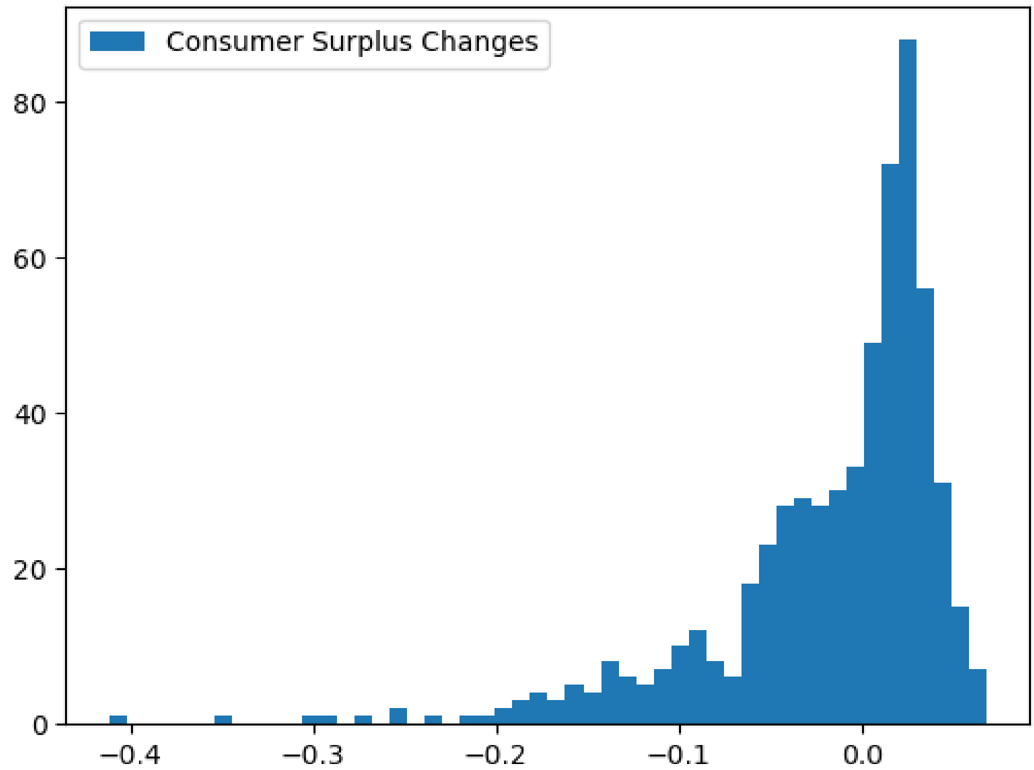

Figure 1: Change in Consumer Surplus

If firms with homogeneous products merge with the marginal costs reduction, the equilibrium price would actually decrease compared to that of pre-merger. The result also suggests an increase in Consumer Surplus, as shown in Figure 1.

6. Explain why this additional assumption (or data on the correct values of Mt) is needed here, whereas up to this point it was without loss to assume Mt = 1. What is the predicted impact of the merger on total welfare?

The per market consumer welfare is calculated as an integration across the consumer base. If we specify the value of M_t wrong, this is essentially leading to a wrong estimates.

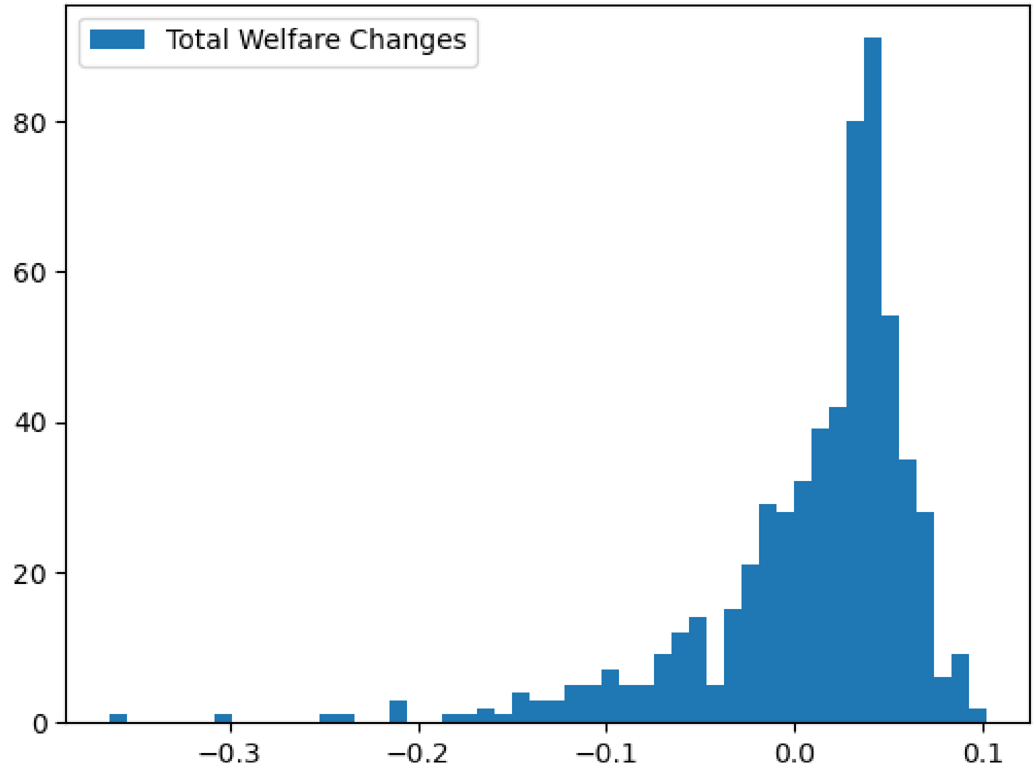

Figure 2: Change in Total Surplus

As shown in the figure 2, there is an increase in total welfare, if assuming the merger of homogeneous product firms merge with a lower marginal cost. The total welfare is calculated as change in the consumer welfare plus the change in firm's profits (aggregating at market-level).

Q2. School Quality and Competition

FOC for p_j

$$P_j = [MC(q_j)] - V] + \frac{S_j(q, p, \xi)}{|\frac{\partial S_j}{\partial P_j}|} \tag{1}$$

FOC for q_j

$$q_j = \left[\beta_j - \frac{V - c_0}{c_1}\right] - \frac{S_j(q, p, \xi)}{\frac{\partial S_j}{\partial q_j}} = 0 \tag{2}$$

Own price p_j and quality q_j

- If $q_j \uparrow$
- MC Channel : $\Rightarrow MC(q_j) \uparrow \Rightarrow P_j \uparrow$
- Market Share Channel: $\Rightarrow S_j(q, p, \xi)|_{\frac{\partial S_j}{\partial q_j} > 0} \uparrow \Rightarrow P_j \uparrow$
- Sensitivity Channel: $\frac{\partial S_j / \partial p_j}{\partial q_j} < 0 \Rightarrow \frac{\partial S_j}{\partial p_j} < 0 \Rightarrow P_j \uparrow$
- **Interpretation (Increase in Own Quality):**

- Own marginal cost increases, leading to more expensive products.
- Own market share increases, more market power, and leading to higher mark-up.
- Own price sensitivity decreases, leading to more mark-up (own price sensitivity regarding the quality is negative).
- **Own price p_j and quality q_j are strategically complements**

Own price p_j and rival price p_k

- If $P_k \uparrow$
- Market Share Channel: $\Rightarrow S_j(q,p,\xi)|_{\frac{\partial S_j}{\partial p_k}>0} \uparrow \Rightarrow P_j \uparrow$
- Sensitivity Channel: $\frac{\partial S_j/\partial p_j}{\partial p_k} < 0 \Rightarrow \frac{\partial S_j}{\partial p_j} < 0 \Rightarrow P_j \uparrow$
- **Interpretation: (Increase in Rival Price)**
  - Own market share increases, more market power, and leading to higher mark-up.
  - Own price sensitivity decrease, leading to higher mark-up (Cross-price sensitivity is negative).
  - **Own price p_j and rival price p_k are strategically complements**

Own quality q_j and rival quality q_k

- If $q_k \uparrow$
- Market Share Channel: $\Rightarrow S_j(q,p,\xi)|_{\frac{\partial S_j}{\partial q_k}<0} \downarrow \Rightarrow q_j \uparrow$
- Sensitivity Channel: $\frac{\partial S_j/\partial q_j}{\partial q_k} > 0 \Rightarrow \frac{\partial S_j}{\partial p_j} > 0 \Rightarrow q_j \uparrow$
- **Interpretation (Increase in Rival Quality):**
  - Own market share decrease, less market power, and leading to higher quality to compete.
  - Own quality sensitivity increases, incentivize on quality improvement (cross-quality sensitivity is positive).
  - **Own quality q_j and rival quality p_k are strategically complements**

Own price p_j and rival price q_k

- If $q_k \uparrow$
- Market Share Channel: $\Rightarrow S_j(q,p,\xi)|_{\frac{\partial S_j}{\partial q_k}<0} \downarrow \Rightarrow p_j \downarrow$
- Sensitivity Channel: $\frac{\partial S_j/\partial p_j}{\partial q_k} > 0 \Rightarrow \frac{\partial S_j}{\partial p_j} > 0 \Rightarrow p_j \downarrow$
- **Interpretation (Increase in Rival Quality):**
  - Own market share decrease, less market power, and leading to lower mark-up.
  - Own price sensitivity increases, leading to less mark-up.
  - **Own price p_j and rival quality q_k are strategically substitute**